

# CAMPUS EXCHANGE

*Student Marketplace Web Application*

|                             |                              |
|-----------------------------|------------------------------|
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| <b>Batch Code</b>           | ANP-D4852                    |
| <b>Project Name</b>         | Campus Exchange              |
| <b>College / University</b> | L.D. Sonawane Degree College |
| <b>Course</b>               | BSC CS 2026                  |
| <b>Project Guide</b>        | Rajshri Thete                |
| <b>Project Type</b>         | Full-Stack Web Application   |

## Project Documentation • Sprint 1

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# 1. Introduction

Campus Exchange is a student-focused marketplace application designed to make it easier for students to buy, sell and exchange academic and college-related items within their student community. The platform provides a centralized interface for discovering listings, communicating with sellers, saving items, creating deals and managing exchange requests.

## 2. Problem Statement

Students frequently need books, notes, calculators, electronics, stationery and laboratory equipment, while other students may have these items available after completing a course. Traditional channels such as informal groups and word-of-mouth make it difficult to search, compare and manage these transactions. Campus Exchange addresses this problem through a dedicated student marketplace.

## 3. Objectives

- Provide a simple platform for students to buy, sell and exchange items.
- Allow users to create and manage their own listings.
- Enable searching, category filtering and product discovery.
- Provide messaging between buyers and sellers.
- Support favorites for items users want to revisit.
- Support purchase Deals with confirmation and meetup stages.
- Support product-to-product Exchange requests.
- Help users discover nearby sellers using location data.

## 4. Proposed Solution

The proposed system combines a React-based frontend with a FastAPI backend. Users authenticate with JWT, browse listings, interact with sellers and manage transactions. SQLAlchemy provides the ORM layer for persistent application data.

## 5. Key Features

| Module         | Description                                                                    |
|----------------|--------------------------------------------------------------------------------|
| Authentication | Registration, login, JWT-based protected actions and logout.                   |
| Dashboard      | Overview of categories and recently added products.                            |
| Browse Items   | Search, categories, filters and listing cards.                                 |
| Create Listing | Students can post products with descriptions, prices, condition and type.      |
| My Listings    | View and manage listings created by the current user.                          |
| Favorites      | Save and remove products from a personal favorites list.                       |
| Messaging      | Communicate with other students about listings.                                |
| Deals          | Manage normal buying transactions through confirmation, meetup and completion. |

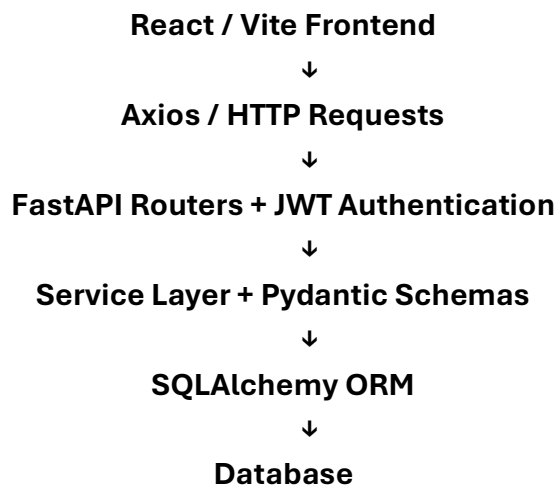
|                       |                                                                |
|-----------------------|----------------------------------------------------------------|
| <b>Exchanges</b>      | Request, accept, reject, complete or cancel product exchanges. |
| <b>Nearby Sellers</b> | Find products from students within a selected search radius.   |
| <b>Profile</b>        | Manage and display student account information.                |

## 6. Technology Stack

| Technology            | Usage                                      |
|-----------------------|--------------------------------------------|
| <b>Frontend</b>       | React, Vite, JavaScript, CSS, React Router |
| <b>HTTP Client</b>    | Axios                                      |
| <b>Backend</b>        | Python, FastAPI                            |
| <b>ORM</b>            | SQLAlchemy                                 |
| <b>Validation</b>     | Pydantic                                   |
| <b>Authentication</b> | JWT / Bearer authentication                |
| <b>Development</b>    | VS Code, npm, Uvicorn                      |

## 7. System Architecture

The application follows a client-server architecture. The React frontend sends HTTP requests to FastAPI endpoints using Axios. The backend authenticates users, validates data through Pydantic schemas, applies business logic through services and persists data through SQLAlchemy.



## 8. Functional Modules

### 8.1 Authentication

Users register and log in. Protected operations require a valid bearer token.

### 8.2 Listings

Users create, view and manage products. Listings include title, description, price, condition, category, type, status and optional location/image data.

### 8.3 Favorites

Authenticated users can save products and later remove them from Favorites.

### 8.4 Messaging

Users can start conversations from a listing and communicate with another student about the product.

### 8.5 Deals

A Deal represents a normal purchase. It can move from Pending to Meetup and then Completed, or be Cancelled.

### 8.6 Exchanges

An Exchange represents a product-for-product transaction. The receiver can accept or reject the request, followed by completion or cancellation.

### 8.7 Nearby Sellers

Latitude and longitude data support location-based discovery within a selected radius.

### 8.8 Profile

The profile area displays account and student information and provides profile-management access.

## 9. User Workflows

### 9.1 Buying Workflow

Browse Items → Listing Details → Buy Now → Deal Created → Buyer/Seller Confirmation → Meetup → Completed → Listing marked Sold

### 9.2 Exchange Workflow

Listing Details → Propose Exchange → Exchange Request → Accept / Reject → Complete / Cancel

### 9.3 Messaging Workflow

Listing Details → Message Seller → Conversation → Buyer ↔ Seller

## 10. Database Design

Important entities include User, Category, Listing, ListingImage, Favorite, Message, Deal, Exchange and Review.



Category → Listings

## 11. Frontend Structure

The frontend is organized into reusable React components, layouts, pages, hooks and service modules.

```
frontend/
├── public/
├── src/
│   ├── assets/
│   ├── components/
│   ├── hooks/
│   ├── layouts/
│   ├── pages/
│   ├── services/
│   ├── App.jsx
│   └── main.jsx
├── package.json
└── vite.config.js
```

## 12. Backend Structure

```
backend/
├── app/
│   ├── models/
│   ├── schemas/
│   ├── services/
│   ├── routers/
│   ├── database/
│   ├── auth.py
│   └── main.py
```

## 13. API Overview

| Resource       | Representative Endpoints                                                                                         |
|----------------|------------------------------------------------------------------------------------------------------------------|
| Listings       | GET /listings/, GET /listings/my, GET /listings/{id}, POST /listings/, PUT /listings/{id}, DELETE /listings/{id} |
| Listing Images | POST /listing-images/, GET /listing-images/{listing_id}, DELETE /listing-images/{image_id}                       |
| Exchanges      | GET /exchanges/, GET /exchanges/{id}, POST /exchanges/, PUT /exchanges/{id}, DELETE /exchanges/{id}              |
| Deals          | GET /deals/, GET /deals/{id}, POST /deals/, PUT /deals/{id}, DELETE /deals/{id}                                  |

## 14. Authentication & Security

- JWT bearer authentication protects authenticated API operations.
- User ownership should be checked before modifying or removing listings.
- Users should not be allowed to purchase their own listings.

- Exchange participants should be validated before changing exchange state.
- Production secrets should be stored in environment variables rather than source code.
- Production deployment should use HTTPS, secure token handling, input validation, rate limiting and logging.

# 15. Screenshots

The following screenshots illustrate the Campus Exchange interface, including the dashboard, marketplace browsing, nearby-seller discovery and related application screens.

## Dashboard

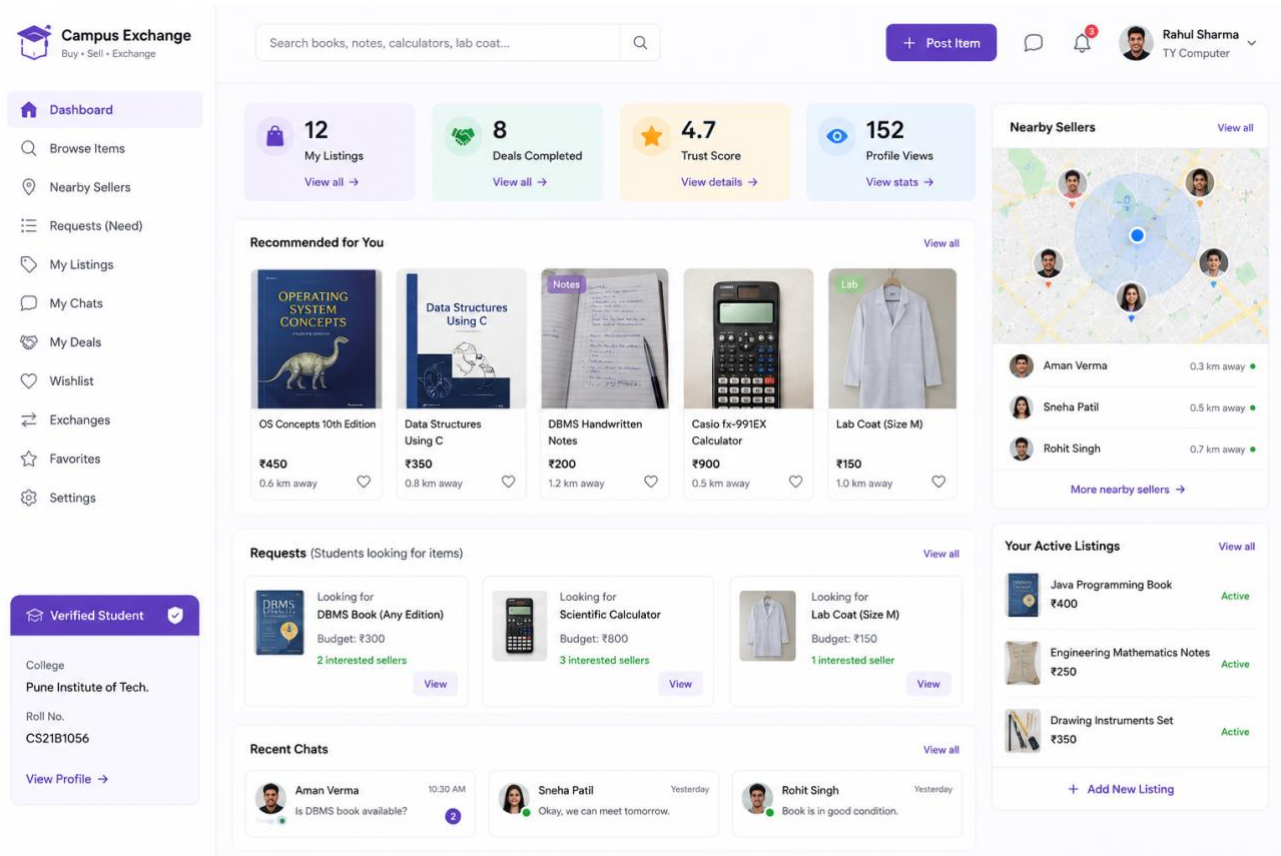


Figure: Dashboard



## Nearby Sellers

**Campus Exchange**  
Buy • Sell • Exchange

Dashboard

Browse Items

Nearby Sellers

Exchanges

Favorites

Settings

Verified Student

College  
Pune Institute of Tech.  
Roll No.  
CS21BI056  
[View Profile](#)

+ Post Item

### Nearby Sellers

Find trusted students selling items near you.

All Items

Books

Notes

Lab Equipment

Electronics

Stationery

Filters

Showing sellers within 5 km

**Aman Verma**  
TY Computer  
Operating System Concepts (10th Edition)  
₹450  
Meet-up: Pune Institute of Tech Gate  
0.3 km away  
Online  
[View Item](#)

**Sneha Patil**  
TY IT  
DBMS Handwritten Notes  
₹200  
Meet-up: College Library  
0.5 km away  
Online  
[View Item](#)

**Rohit Singh**  
TY Computer  
Casio fx-991EX Calculator  
₹900  
Meet-up: Main Canteen  
0.7 km away  
Online  
[View Item](#)

**Kavya Joshi**  
TY IT  
Lab Coat (Size M)  
₹150  
Meet-up: College Gate  
1.2 km away  
Online  
[View Item](#)

Load More Sellers

Your Location  
Pune Institute of Tech.

You

Online  
Active Now

1 km  
Very Near

3 km  
Nearby

5 km  
Farther

**Meet Safely**  
Always meet in public places like college gates, library or canteen.

[Read Safety Tips](#)

Figure: Nearby Sellers

# Exchanges



- Dashboard
- Browse Items
- Nearby Sellers
- Exchanges**
- Favorites
- Settings

Verified Student

College  
Pune Institute of Tech.  
Roll No.  
CS21B1056  
[View Profile](#)

Search books, notes, lab equipment and more...

+ Post Item



Rahul Sharma  
TY Computer

## Exchanges

Exchange your items with other students. Save money and help each other!

- All Exchanges**
- My Offered Items
- My Wanted Items
- My Exchanges

How Exchange Works?

Filters [Clear All](#)

### I want (Category)

All Categories

### Offered Category

All Categories

### Distance

- ☐ Within 1 km
- ☒ Within 3 km
- ☐ Within 5 km
- ☐ Within 10 km

### Availability

- ☒ Any Time
- ☐ Today
- ☐ This Week
- ☐ This Month

[Apply Filters](#)

Showing 8 exchange opportunities

Sort by: **Newest First**

|                                                               |                                                                                                      |  |                                                                                       |                              |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------|------------------------------|
| <br><b>Aman Verma</b><br>TY Computer<br>★ 4.8<br>0.3 km away  | <b>Offering</b><br><br>Operating System Concepts (10th Edition)<br>Good Condition<br>Offers by: Aman |  | <b>Wanting</b><br><br>Data Structures Using C<br>Good Condition<br>Offers by: You     | <a href="#">View Details</a> |
| <br><b>Sneha Patil</b><br>TY IT<br>★ 4.7<br>0.5 km away       | <b>Offering</b><br><br>DBMS Handwritten Notes<br>Like New<br>Offers by: Sneha                        |  | <b>Wanting</b><br><br>Java The Complete Reference<br>Good Condition<br>Offers by: You | <a href="#">View Details</a> |
| <br><b>Rohit Singh</b><br>TY Computer<br>★ 4.6<br>0.7 km away | <b>Offering</b><br><br>Casio fx-991EX Calculator<br>Like New<br>Offers by: Rohit                     |  | <b>Wanting</b><br><br>Lab Coat (Size M)<br>Good Condition<br>Offers by: You           | <a href="#">View Details</a> |
| <br><b>Kavya Joshi</b><br>SY IT<br>★ 4.9<br>1.1 km away       | <b>Offering</b><br><br>Engineering Mathematics Notes<br>Good Condition<br>Offers by: Kavya           |  | <b>Wanting</b><br><br>Drawing Instruments Set<br>Like New<br>Offers by: You           | <a href="#">View Details</a> |



### Why exchange?

- ☒ Save Money
- ☒ Reduce Waste
- ☒ Help Fellow Students
- ☒ Build Connections

+ Post an Item to Exchange

Figure: Exchanges

## Dashboard / Recommendations



### Dashboard

- Browse Items
- Nearby Sellers
- Exchanges
- Favorites
- Settings

### Verified Student

College  
Pune Institute of Tech.  
Roll No.  
CS21B1056  
[View Profile →](#)

Search books, notes, calculators, lab coat...



+ Post Item



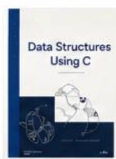
Rahul Sharma  
TY Computer

### Recommended for You

[View all](#)



OS Concepts 10th Edition  
₹450  
0.6 km away



Data Structures Using C  
₹350  
0.8 km away



DBMS Handwritten Notes  
₹200  
1.2 km away



Casio fx-991EX Calculator  
₹900  
0.5 km away



Lab Coat (Size M)  
₹150  
1.0 km away

### Popular Categories



Books  
128 items



Notes  
96 items



Lab Equipment  
45 items



Electronics  
38 items



Stationery  
62 items



Others  
54 items

### Recently Added

[View all](#)



Java The Complete Reference  
₹400  
0.4 km away



Engineering Mathematics Notes  
₹250  
0.9 km away



Physics Lab Kit  
₹1,200  
1.1 km away



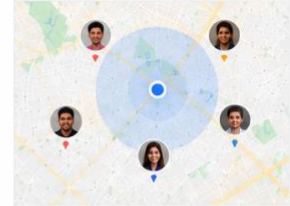
DSA Handwritten Notes  
₹300  
1.3 km away



Scientific Calculator  
₹750  
0.6 km away

### Nearby Sellers

[View all](#)



- Aman Verma 0.3 km away
- Sneha Patil 0.5 km away
- Rohit Singh 0.7 km away

[More nearby sellers →](#)

### Trending Items

[View all](#)

- Python Programming Book  
₹350  
24 interested
- Operating System Notes  
₹200  
18 interested
- Lab Coat (Size L)  
₹150  
15 interested

[Explore more items →](#)

Figure: Dashboard / Recommendations

## Marketplace



- Dashboard
- Browse Items
- Nearby Sellers
- Exchanges
- Favorites**
- Settings

Verified Student

Pune Institute of Tech.

Roll No.  
CS21B1056

[View Profile](#)

Search books, notes, calculators, lab coats...

+ Post Item



**Rahul Sharma**  
TY Computer

### ☆ Favorites

Items you have saved for later.

Sort by: Recently Added

All (12) Books (6) Notes (3) Lab Equipment (1) Electronics (1) Stationery (1)

Operating System Concepts (10th Edition)  
₹450  
0.6 km away

DBMS Handwritten Notes  
₹200  
1.2 km away

Casio fx-991EX Classwiz Calculator  
₹900  
0.5 km away

Lab Coat (Size M)  
₹150  
1.0 km away

Data Structures Using C (2nd Edition)  
₹350  
0.8 km away

DSA Handwritten Notes  
₹300  
1.3 km away

Java The Complete Reference  
₹400  
0.4 km away

Engineering Mathematics Notes  
₹250  
0.9 km away

Physics Lab Kit  
₹1,200  
1.1 km away

Drawing Instruments Set  
₹350  
0.7 km away

Scientific Calculator (fx-991CW)  
₹750  
0.6 km away

Python Handwritten Notes  
₹250  
1.2 km away

Showing 12 of 12 items



Don't see an item you favored?

Some items may have been sold or removed by the seller.

[Browse Similar Items](#)

Figure: Marketplace

## 16. Testing Checklist

- ☐ Register with valid details
- ☐ Login with valid and invalid credentials
- ☐ Verify protected pages/actions
- ☐ Create and view a listing
- ☐ Remove an owned listing
- ☐ Prevent unauthorized listing modification
- ☐ Add and remove Favorites
- ☐ Open and send Messages
- ☐ Create and confirm a Deal
- ☐ Cancel and complete a Deal
- ☐ Create, accept and reject an Exchange

- ☐ Complete or cancel an Exchange
- ☐ Test Nearby Sellers with location access
- ☐ Verify profile information

## 17. Future Scope

- Real-time chat using WebSockets
- Push/in-app notifications
- Browser geolocation and map integration
- Advanced search, sorting, filters and pagination
- Cloud image storage
- Email verification and password reset
- Admin dashboard and listing moderation
- Improved transaction-based review system
- Automated unit and integration testing
- Production deployment and monitoring

## 18. Conclusion

Campus Exchange provides a structured digital marketplace for students to discover, sell, buy and exchange useful academic and college-related products. The project combines a modern React interface with a FastAPI backend, JWT authentication and SQLAlchemy-based data management. Its modular design provides a solid foundation for future features such as real-time communication, maps, notifications, moderation and production deployment.

## Student & Project Details

|                             |                              |
|-----------------------------|------------------------------|
| <b>Student Name</b>         | Sujal Rai                    |
| <b>Student ID</b>           | AF05215292                   |
| <b>Batch Code</b>           | ANP-D4852                    |
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| <b>Project Type</b>         | Full-Stack Web Application   |